This is your full case study on Railcom which will form the basis for your presentation of recommendations. It is a big case study with lots of different issues; your job as a team is to focus in your pairs on the KEY problems and to research and present on the solutions and ongoing recommendations. Take time to read this through carefully and thoroughly, and highlight the issues of People, Processes and Technology. Remember CONTEXT is the key to good marks! Good luck with your mission!

Railcom: a railway disaster?

Taking Stock

The next day Hank came by Alice's office. She was tired and uneasy. Alice wanted to do her job well, but she wasn't sure how she could if no one saw the importance in what she was saying. "Alice," said Hank. "Let's talk about this cyber "stuff" you mentioned. You know, maybe we can find a little bit in the budget, and we could do some minor changes."

Alice looked up at him. "A compromise?" She spoke. "I'm happy to see that you've thought about it. I'm trying to do my job, Hank. I've been reading in the news lately that all these companies abroad have been getting hacked, money and information stolen, and then they don't know how to clean up the mess. I don't want that to be us."

"You're right," Hank said with a sigh. "I just read in the news this morning that another company was hacked with some ransomware, it cost that company millions to fix everything. So, let's put a little money in and then we can advertise to our customers that we just updated our systems and now we are more secure than ever before."

"I like the idea. But what you're proposing is simply the tip of the iceberg. Throwing money at the problem is like trying to stop a haemorrhage with a Band-Aid. Besides, we can't afford not to be protected. Both of us know that a budget is not the same as changing the company culture and mindset. We need to upgrade some of our system, but we need to reengineer our processes and educate our people too. We need to fix all aspects of the problem: people, processes, and technology. We can put money to tweak our systems, but when a good system encounters a bad culture, the bad culture wins. Our processes are old, our people, including our suppliers' employees and our customers, are not up to date with the changing environment. Plus, our technology is a mismatch of legacy and new systems threaded together. It won't protect us for too much longer. Adding a small amount to the budget will not solve it, Hank."

Alice paused. She needed to break it down for Hank. To get him to fully understand that there were more weaknesses in Railcom 2.0 than can be addressed with a few system updates. "Think about these three aspects, Hank. I can update the systems with a little bit of money, but this needs to be a company-wide change. Operations, marketing, sales, HR, and IT need to get on board."

Hank pulled up a chair. He had no doubts about Alice's competence and wanted her to get her point across. "Nobody questions your calibre or intentions, Alice. But yesterday's meeting might not have been the best venue for expressing your concerns. That too, right after James' vision of Railcom 2.0. So, why don't you break it down to me the specific issues? I can then take it back to James and see what can be sorted."

Alice felt a sense of ease wash away her exhaustion. Finally, she had Hank's ear. Alice took her tablet out and went on to explain.

"We need to think about technology, processes, and people."

"Our technology at Railcom needs updating, we are all aware of that, even our customers know that. The technology is a company problem, but it makes us vulnerable both internally and externally. We have to get it fixed. Railcom uses a combination of old systems and databases, not to mention out of date firewalls.

We have more of a craft project than we do an actual system. Anyone with the slightest know-how could easily take advantage of all the holes in our system. You and James need to realize that you can't always just buy the cheapest system or cheapest protection and assume all the pieces and parts will fit together. It doesn't work like that. A hacker could easily detect the smallest hole in our systems, and we probably wouldn't have a clue! Not to mention our lack of logical separation in our systems – that makes us highly susceptible to a full-scale Ransomware attack. Remember the NHS attack? Well, we could be next!"

She stopped to look at Hank. He seemed a bit dumb founded. I don't think he quite understood the issue with the mismatched systems. But she knew that wasn't the end of vulnerabilities for Railcom. She decided to continue. You can't only protect one corner of the triangle and leave the other two vulnerable; the triangle will crumble.

"Then comes people. Our people, Hank, that's another worry of mine. Fixing this issue isn't as easy as spending some money for system upgrades and repairs. This will take time and education and commitment from all divisions of the company. I walk into the IT department and guess what I see! Passwords written on paper stuck behind monitors or under keyboards, people logging in and then leaving the computers unattended. Not to mention how many other employees use their personal cell phones to do work for our company. All of these activities put Railcom at risk. Our people don't understand the importance of confidentiality, nor of security. Would our employees be able to tell the difference between an email from James or a fake email that looks like it's from James? Our employees can quickly become a vulnerability for Railcom since their behaviour basically invites someone to hack us. But you know, a lot of the problem is with James and the company culture itself. We can't expect people to understand the importance of security if we don't promote it within the company culture first."

Alice knew that training employees would take time. Even if Railcom were to update its systems, it wouldn't do the company any good if the systems were left unprotected by its own employees unaware of the lurking risks. Employees are often the quickest way to breach a system. Every employee creates a potential risk for Railcom.

Alice cautiously noticed Hank's reaction. He hadn't drifted away. After a momentary pause, Alice continued.

"We need to also focus on Railcom's processes, Hank. I've only been with the company for a short while and things were a bit of a cybersecurity mess before me, but I am working on fixing them. My next project is going to tackle some of the processes in the company. Do you know that as of now, employees are not prompted to change their password nor create strong passwords? This is basic security, a simple process fix.

In addition, when employees log onto their systems, there is no other requirement than just the password. We really need to add a few more layers of protection, something that would increase the complexity so it wouldn't be an easy breach for a hacker. But we can't start implementing a total process revamp unless our people understand the importance of it. The last time we ran an initial systems audit; we found that there is little separation of

duties. The programmers are often rotated as testers and even database developers. Without any formalized process for security demarcation, any one of them can easily create vulnerabilities. Then we have our authentication process - I mean, what authentication process? I know of multiple instances where one of our salespeople called IT about a link not working. It was a third-party link, and IT ran the URL (link) without verification. What if the link carried malware? And we don't even have a proper incident response plan in place. It could have been a disaster in the making."

Alice stopped. She had made her point. She looked at Hank for his take on her concerns.

Hank felt overwhelmed from the information deluge. Yet, he appreciated every point that Alice made. At the end of the day, a breach could turn Railcom 2.0 upside down and open up a litigious can of worms. The cost could be prohibitive.

"Well Alice," Hank deliberately leaned forward to convey his sincerity. "It seems like you have a full report of the company. That's great, that's what we want you to do. True, we need to do something about them now, but that needs getting operations and HR together at the table. That might take a few months, given rollout pressures. But I promise you, I'll get it done. Let's make this our next priority with a six-month timeline for deployment."

Alice felt restless once again. "Hank, it's not that simple. The rollout makes Railcom 2.0 open to the public. Every hacker and half-baked script-kiddie will roll up his or her sleeves to penetrate our systems. It's a game to them, sometimes more malevolent than just a game. A six-month timeline is too long a horizon in an age when the mice get smarter than mice-traps in seconds. Rolling out Railcom 2.0 with cybersecurity on the backburner is not wise. I know James wants to create a competitive advantage with a quick rollout, but I'd rather be careful."

Hank and Alice ended the meeting with the compromise to update the systems with the budget Hank could spare. She wasn't pleased but it was a start. Once at her desk, she emailed James about a cybersecurity workshop with HR and the IT department. It would be of the utmost importance to ensure that all the employees begin some sort of workshop on cybersecurity sooner rather than later.

It was within this landscape that Railcom was operating two divisions. The railway system division handled internal operations including timetabling, rail services, employee payroll, among others. The customer service division mainly focused on ticketing, access, and customer relationship management. Systems across the two divisions were loosely connected by middleware that seemed more like an afterthought than a deliberation. Railcom needed to stay competitive in a time when railroads appeared to be a thing of the past. In its drive to keep up with the times, Railcom used its IT group to tackle all technology requirements across both divisions. The same IT personnel that were developing the CRM database could, in a few hops, also access ticketing information. There was no separation of duties.

Railcom had just launched the ComCard. Much like the Oyster Card in the U.K., ComCard would allow prepaid multi-transit advantages to customers. Sadly, ComCard was only sold at the ticket counters or newsstands. Railcom announced that the ComCard would be available for purchase online in a few months. Railcom was going to embrace the digital era. What could go wrong?

A Curious Unravelling

It all began with a seemingly basic call from payroll. For some reason, payroll funds were short, and HR wanted to know if it was a glitch in the system. Alice was confused by the phone call but figured she would personally follow up with the IT department just to be safe. She reached out to her IT Director, Lev.

Lev responded within the next five minutes. All systems were good, and validation checks were in place: no glitches, except an unknown stream of transactions that appeared to be cross-account transfers. But when he looked closer into the transfer schedule, it seemed suspect. Someone was siphoning funds to an external account.

Alice paused with a sense of dread. She understood all too well what had happened. Railcom 2.0 had just been compromised. Funds had been accessed and stolen through the payroll system.

Before Alice recovered from Lev's response, she received another email from the customer support department. Strange, she thought, what could customer support want from her at this hour?

"Shouldn't customer service be escalating all matters to the supervisor and then to Lev?" Alice thought? "Who knows?" she said out loud as she shook her head. There was no established escalation and incident response plan in place. She felt as if she was facing a typhoon in a tattered raft.

The customer service email signalled panic. Hundreds of customers were starting to complain that they had ordered a ComCard from the website but hadn't yet received it, even though the customers had been charged. She tried to consider all the possibilities; did James launch the online ComCards already, without telling her? No, couldn't be. She searched for the website; she wanted to see it herself. Before she had finished reading Lev's response to the payroll funds shortage, Alice found herself feverishly forwarding Lev the latest ComCard crisis.

She opened the email and began to read. The more she read, the quicker she read. Customer support had forwarded her a series of customer emails, all with the same heading "Purchase your Railcom Tickets here." The email address, ticketsales@railcoms.com, appeared credible except for a minute detail; the domain, instead of railcom.com, read railcoms.com. Below it was a well-crafted message:

To all valued customers,

Railcom announces its brand-new online ticketing service, Railcom 2.0. We invite you to purchase a ComCard as the next-generation ticket and save 15% off our regular fares. This special offer will end by midnight, the 17th of June, 2024

Please click this special link at www.randcardsforrailcom.com to purchase the ComCard and activate your discount.

Sincerely,

James Rith, CEO

Railcom

She cautiously hovered over the www.comcardsforrailcom.com web address and looked at the real web link. It was an open IP address, 107.xx.xxx.xxx. She needed to see where the click would carry her. Alice made sure her security was active, and all virus and malware signatures were up to date before proceeding. She clicked. A site, phished and spoofed to look like Railcom, opened up. Simultaneously she did her application security. The site was trying to execute a Trojan bot onto her computer. This type of Trojan bot worked independently and secretly collected financial information and transferred it to a remote server controlled by criminals.

Alice felt a cold sweat. How many customers have fallen for this? She thought. She took a deep breath; Railcom had just been hit on two fronts in such a short span of time.

She needed to tell the board. Alice sent an email to James requesting an immediate conference call and telling him briefly about the crises. Within minutes James responded, suggesting a conference call in 15 min. It was 2 am.

As Alice dialled into the conference call, her mind was racing. When was payroll breached? What else might have been compromised? Railcom's R&D, release data, the lot? When did the fake website get built? The timing was too uncanny. Were the two events connected? How much damage will it do to Railcom's customers and the Railcom brand?

Her head was spinning. It wasn't even 2 months ago that they updated their systems, added new software, and patched up a few vulnerabilities on the website. James and Hank gave her the budget to fix some immediate issues, and yet her worst nightmare was now a reality.

The phone was ringing now. With each ring, Alice prepped herself about Railcom's system updates. The website was secure, firewalls were in place, and some intrusion detection for the application and content servers. All essential operating software had been updated. That had taken up the budget. Process reengineering and training were budgeted for the six-month horizon. She joined the call.

"Hi. Yes, its Alice. Dialling in for the meeting. Where are we? Any new updates?" Alice felt nervous.

James was frantic. "Alice, tell me what happened? I thought we put money into our systems just 2 months ago! How the hell did this happen? Do we even know what they took? Should we tell the customers or keep quiet?" Railcom 2.0 was meant to be a game changer for the company and for the people, so how could something bad happen to something meant to do good? It just didn't make sense, James thought.

Alice took a deep breath. "Yes. Hi James. I've been up all night. We implemented new software, made sure it was all up to date. Our payroll information resides behind our firewalls, not in the DMZ, you know, the publicly accessible demilitarized zone. So, it couldn't have been a total compromise of our systems. There is just no way for someone to externally get into our systems." Externally. That was the word—externally. Alice's mind stopped racing and became more focused.

"James, remember I suggested a cybersecurity training program for our employees? I know you thought that a system update would be a one-time fix. But, what about our employees? Do they understand the importance of the data we handle? Do they know how to keep our systems secure internally, from our end? And do we vet our employees? They are the ones

that are our 'trust interface' with the public. Do we have the right processes in place to ensure trust?" Alice started to think of all the possibilities. People are always the weakest link.

James sat back in his chair. He did tell HR to do something about a cybersecurity workshop. But really, what wouldn't the employees already know? Just don't give our company information; it's simple, he thought. "Well, Alice. HR and IT were expected to schedule cybersecurity workshops and training for the employees. But the Railcom 2.0 rollout was priority. And really, Alice, do you think our employees wouldn't know how to be secure?"

Alice felt her shoulders tense and her knees lock. Cybersecurity is part of a company culture. If the CEO does not appear to care about championing cybersecurity, why should the employees care? The lax company culture about cybersecurity went all the way to the top. James had good intentions but was too complacent and trusting about the new threat landscape.

She raised her head and spoke into the phone, "James, most threats nowadays originate from inside companies, whether the employee intentionally or unintentionally acts. A culture of being unaware is still as big of a threat as being malicious. If our own employees are unaware of cyberthreats and vulnerabilities, then they can end up putting the whole company and our customers at risk." She wondered why she was the only one considering this possibility.

James's voice broke through her thoughts. "You know, Alice. I trust my people. I don't think any of them would intentionally harm this company. And I can't imagine anyone wanting to target Railcom. I'm sure this is all a system malfunction or something. Just a glitch! Think about it, we aren't a bank or a healthcare company, we are a train company; what valuable information could we possibly have?"

"James." Alice said. "Lev, our IT director just emailed me the logs. It is not a system glitch. It was an intrusion. Perhaps a malware inadvertently installed by our employees. Lev is doing a complete sweep of all terminals and systems as we speak, and we'll know something soon. We ended up installing a bunch of new technology and software to speed up the rollout, but we never changed the processes. With such a mishmash of software and systems, we don't even have a formal process for patch updates and point-to-point encryption because an update to one piece of software could end up changing config files and other parts of the system could slow down or stop working. We don't even have a formal risk management process or even a process or incident response plan (IRP) for incidents like this." Alice paused.

Alice looked down to her computer screen. Lev's sweep report had just arrived. Multiple terminals were infected with a Trojan virus—the same Trojan virus. Railcom had always practiced file-sharing to increase collaboration, and the file-sharing could have allowed the malware to infect multiple computers. Alice took a quick stock and returned to the call.

"James, this is what we know. Some kind of malware infected our employee terminals, and the file sharing service made matters worse. We think it is a keylogging malware, one that logs all the keystrokes and then sends the information back to the cybercriminals. That could have compromised User ID and passwords. Lev is taking care of the issue, but the harm is done. We know that someone was able to obtain employee passwords and information in order to infiltrate our system and siphon money from payroll." Alice paused

for a moment. She was exhausted, but the drive to know exactly what happened kept her going.

Alice continued describing the situation in a controlled manner. "Also, we are aware that there was a fake website registered to an unknown party under the name comcards.com that was selling our ComCard online as the latest offering from Railcom 2.0. Now we all know that the ComCard is currently available only at physical ticket counters. This spoofed website used a phishing attack to make people believe that Random 2.0 had just launched and that people could purchase ComCards directly from the phished website. We also know that this website managed to operate for 1 month without our knowledge, thus successfully stealing customer information and payment. I ran a WHOIS to find out where the attack and the IP originated but it looks like some anonymously registered overseas entity. But the perpetrators could have masked it. They could be anywhere."

Alice's voice now turned resolute. "What we don't know is whether or not these incidents were separate or orchestrated. We are calling in the federal branch. We have to figure out a way to respond to customers who purchased ComCards from the phished website. We have to find a way to communicate to our employees the payroll incident and how we are addressing it. Most of all, we have to address the loss of trust in Railcom to the public. How did this happen, like a perfect storm?"

Alice took a deep breath. Even over the conference line, an anxious, eerie silence was palpable.

Hacking, She Wrote

One month after the minor Railcom system update, in a quiet corner of an Internet cafe a few blocks from Railcom's corporate offices, Kira felt uneasy with excitement. The last company she hacked had gone well, or so she had thought. She was able to take enough money to last her awhile. But this time, it was different.

Railcom had been heavily marketing Railcom 2.0, its new online train scheduling and ticketing system. People were talking about it and anticipating its release. Kira knew Railcom had spent millions creating the new system; it was all over the news. New systems have learning curves and vulnerabilities, she knew that. Three-pronged vulnerabilities: technology, people, and processes. Kira knew that it was easier to leverage vulnerabilities that were more acute than just technology—vulnerabilities in people and processes. And Kira knew that people were always the weakest spear in the cybersecurity triangle. She was going to go for it. Over a latte and a croissant.

Kira reminisced about her last attack. A few years ago, she had managed to siphon millions from a national bank, one that touted its "secure and safe" system to its customers. She had managed to penetrate the bank's system and maintain access for 3 days while the bank was closed for the holidays. It was a simple hack. Once she was inside the bank's systems, all she had to do was set up a few fake accounts at the bank with fake IDs. On the day of the cyber-robbery, she increased the withdrawal limit on these accounts from inside the bank's system and then transferred the money to external accounts at other banks. It was so simple, she mused. She had set up at another Internet cafe on the westside of town and was able to complete the heist while having her macchiato and scone, different tastes for different heists.

Kira's latest inspiration came from reading an article about international cybersecurity state-of-affairs. "The majority of cyber-crime, even the most sophisticated, is based on the simple exploitation of traditional IT access credentials – cards, PINs, passwords (CPPs) – it's just so easy."

Kira, nonetheless, felt uneasy and on edge. She was an employee at Railcom. It was unethical but Kira had lost her ethical bearings a long time ago. It was risky, but she might be able to hide her crime with a twin-flank attack. She had to set the stage.

Her first attack flank would come from outside the company. She had set up a fake website that lured customers to give personal information and credit card information in return for a railroad card sent via mail within 30 days. This would buy her a few weeks before customer complaints would flood the company. In addition, she was taking advantage of the fact that Railcom had announced it would make the ComCard available for online purchase, but did not specify a date. Unwary customers were easy to fool.

As for the internal attack, she would go through the payroll system. Payroll was a legacy system, one that had rarely been upgraded or updated. With the newfound excitement about the Railcom 2.0 platform release, she assumed that no one in the company would even monitor the payroll software. The company spent a lot of money on firewalls and some IDS (intrusion detection systems) but mainly for services surrounding Railcom 2.0. Payroll would be "the" candidate for the breach. All she needed to do was use the spoofed email she used for her fake website and send payroll employees a credible "memo" email asking them to open an attachment hiding a keylogger malware. As long as the memo subject read

something mentioning "Railcom 2.0," employees would open it with all urgency and little suspicion. People were always the easiest way to access any system, especially the Railcom employees.

Kira knew that every company falls prey to the same false notion: that technology is their only cybersecurity vulnerability. She would leverage her attack on three fronts: technology, people, and processes. Three points of weakness in a company and Railcom was no exception.

Technology

First, Kira knew that Railcom's systems were a product of evolution from the 1980s. There were legacy systems everywhere. What worked then was still in use, despite Alice's minor upgrades. New software was band-aided to the legacy patchwork with some middleware and most advanced software was on the server-side. Client-side applications were often outsourced with more emphasis on visual appeal at the presentation layer rather than ensuring resilience. Thus, it was what came to be a hodgepodge of different platforms, software, and web services. One didn't even have to be an employee to know this information; it was public knowledge, and a quick Internet search revealed a litany of technology issues Railcom had suffered over time.

Alice, the CIO, had recently been appointed and she wanted to revamp the entire architecture based on a robust, six-sigma software service management agenda. Alice wanted all legacy systems to be moved server-side with better access control and software-as-a-service. Alice's vision was well known and lauded by business and IT magazines. The change was coming in the next year or so. For a hacker, time was of the essence.

With the announcement of Railcom 2.0, Kira knew that Alice could merely focus on minor software and patch updates. Railcom had recently been touting its new system upgrades for Railcom 2.0 to its customers. But overemphasizing Railcom 2.0 had left Railcom's legacy systems unprotected. As an employee, it was easy for Kira to know about vulnerabilities in the payroll system. Companies often think they need protection from external attacks and skimp on protecting themselves against internal attacks, as well. This was Kira's chance.

People

Whether it is a company or a country, people are the weakest link in the cybersecurity triangle. Railcom's executives barely championed cybersecurity as a part of the company culture. Cybersecurity was an IT buzzword. Even as an employee, Kira knew of Railcom's little to no attention given to cybersecurity. Managers didn't care if employees shared passwords or left them written in the open. There was no clean desk policy. Employees could print any document and leave it in the open or copy it on USB devices. There were a few policies, but no-one cared about implementing or enforcing them. No-one cared if a personal phone or laptop was used for work purposes, especially because it saved the company money. And of course, Railcom had never run formal company-wide cybersecurity and risk training on even something as simple as a phishing scam. Even in a digital world, Railcom never educated its people on the dangers and risks to data as a corporate asset. Employee ignorance, phishing emails, and malware: that would be Kira's recipe, and employees would be her channel for malfeasance.

Processes

Kira had heard one too many times an old saying: "Combine good technology, good people, and bad processes. Bad processes win." She knew that Railcom had rarely changed its processes in decades. There was no separation of duties. A payroll employee could also be assigned to help with building an employee benefits database with full access. Railcom interns were rotated across multiple departments, but their old access privileges were never deactivated. That meant that an intern could access multiple different roles at any point of time.

Railcom had a cybersecurity process policy on paper: predict, prevent, detect, and respond. Predict risks and vulnerabilities by periodic systems audits; prevent cyber-attacks by ensuring every system and terminal was secure; detect threats with proactive monitoring; and respond to attacks in a systematic way. But Railcom never implemented them. Frankly, Kira was certain that not even 10 percent of the company knew of Railcom's cybersecurity process or the policy. If Railcom were hacked, a lack of legacy system monitoring would let an attacker roam free for days before being noticed.

Finally, Kira knew the process employees used to log on to their accounts, a simple password verification. It was easy really. She couldn't even remember the last time Railcom requested password changes or data encryption. Most companies don't realize that the process itself is a form of security as well, especially internal processes. All she had to do was obtain a few passwords and she'd be in, nothing more than that. Nothing was encrypted so accessing data, including financial data, would be simple. It was a process waiting to be exploited.

Facing the Fray

Dawn crept through the concrete and trees, glistening the glass panes at Railcom's headquarters. Alice had stayed awake that night, piecing the puzzle together. As morning crept in, it was starting to make sense. For cybersecurity to work, the cybersecurity triangle of technology, people, and processes must work in unison. Even if the system was updated and secure, what happened if processes weren't well defined, and employees weren't educated? A laissez-faire corporate culture towards cybersecurity does not help. Cybersecurity cannot be blind to unseen risks.

The answer, Alice realized, lies in taking a proactive attitude towards cybersecurity. "When you invent a new mousetrap, mice get smarter," Alice dryly mused. Railcom needs to be proactive across all fronts: technology, people, and processes. Technological proactivity needed a strong asset management and monitoring culture. Every system needed protection and continuous monitoring. From a people perspective, Railcom needed to indoctrinate in its employees and customers a culture of prevention and accountability. Finally, Railcom needed to reengineer its processes to creep out of antiquated ways of managing assets, auditing systems, preventing breaches, and responding to incidents.

But for now, first things first: How should Railcom communicate the two breaches to its employees and to the public? Some of, if not all of, Railcom's employees are also Railcom customers. That added to the complexity. Communicating the breaches was imperative. Yet, she was unsure about how to proceed. Should she wait for the Investigators to brief her? Or should Railcom address it right then and there, in the open? What would be the fallout of each choice?

Perhaps this incident will be an eye opener for James. He held everyone's respect, and his cybersecurity championship might finally start changing the company culture. But, what else? Who should educate the customers and change the customer culture? Railcom needed a national culture, a proactive and preventative attitude towards cyber-awareness. Will this be a wake-up call for Railcom to create a more engaged culture of cybersecurity awareness, accountability, and enforcement?

As Alice heated some stale coffee in a microwave, a new email from Lev arrived. Lev had been looking at various web, access, and query logs to locate anything that could shed more

light on the two incidents. Lev's initial assessment confirmed Alice's original suspicion. The fake website used to steal customer payments and the internal breach to steal from the payroll system were linked. It had to have been an employee that was familiar with the company, familiar with the software, and familiar with the general company culture about cybersecurity.

Lev's email reported that he had heard that a new email was beginning to circulate, aimed at Railcom employees and customers. The email was another attempt at social engineering. The email was spoofed to look like a Railcom email. It was veiled as a update on the breach and asked all email recipients to click on another link, certainly to a phished website, to update their passwords for safekeeping. That would put more employee and customer information at risk. Alice had forgotten to sip her coffee. She scrambled to craft another response. It was never-ending.

Alice felt helpless. She was facing one or more perpetrator or perpetrators devoid of ethics and careless. There are anonymous, nameless, faceless entities lurking in virtual shadows—intent on causing mayhem and malfeasance. It was criminal. They left a path of destruction but few footprints. Where should she focus? Stop customers from purchasing fraudulent ComCards from the phished site? Isolate and fix the payroll system without knowing what else may have been compromised? Address the new social engineering crisis that could further compromise employee and customer personal and financial information?

One thing was certain. The Railcom brand had been tarnished for a long time to come. The damage to Railcom was more than money; it was trust. Railcom stood to lose its customers, its reputation, and its credit rating. This could invite a stream of litigation. Alice knew that she needed to face the fray and craft a plan of action. She was always resolute, even in the face of adversity. She closed her eyes and breathed deeply.

As Alice emailed the executive team to schedule an urgent meeting in the next hour, her video intercom buzzed. Digital forensic investigators had arrived to help. Alice readied herself for the first of many meetings and many a sleepless night ahead.

Over to you!